GENERAL SPECIFICATION G3.2

Site Investigation Works

CONTENTS

G3.2 SITE INVESTIGATION WORKS

G3.2.1 Scope

(a) This Section contains requirements which, where relevant to this Contract, shall apply to site investigation works as detailed in the specification and where required by the S.O. to determine geotechnical conditions. Site investigation shall include, but not be limited to including light cable percussion boring and in-situ testing in order to determine the nature and sequence of the sub-surface strata, their material properties and ground water conditions as required.

G3.2.2 Extent of the Site Investigation Works

(a) In particular, the site investigation of the Works includes:

(i) Boreholes in soils and soft strata for the preparation of logs and to permit in-situ tests to be carried out;

(ii) Standard penetration tests and in-situ vane shear tests;

(iii) Groundwater monitoring and instrumentation;

(iv) Determination of the positions and levels of all investigation holes;

(v) Submission of daily records of field work;

(vi) Preparation of draft logs and of preliminary copies of the results of all field tests; and

(vii) Submission of draft and final copies of bound factual reports containing details of the method of working, final logs and results of field tests.

G3.2.3 Standards and Codes of Practice

(a) Unless otherwise specified, the Works included in the site investigation shall be carried out in accordance with BS EN 1997-2.

G3.2.4 Position of the Site Investigation Works

(a) The S.O. shall instruct the exact location at which the exploratory holes are required during the course of the Contract. The positions and levels of these holes shall be set out on site by the Contractor. These positions may be varied by the S.O. as the work proceeds in the light of the results obtained and other factors. Before work is started at any hole the Contractor shall confirm its position and probable required depth with the S.O.. These details shall be recorded either in the form of a written instruction from the S.O. to the Contractor, or if this is not received, by written confirmation from the Contractor to the S.O. before work commences at the hole location.

(b) As soon as possible after work has begun at any exploratory hole location and before its completion the Contractor shall accurately survey its location and determine and its ground level from which its depth measurements are logged.

G3.2.5 Order and Timing of Site Investigation Works

(a) The S.O. may direct in what order the exploratory holes shall be made.

(b) The required timing of the site investigation component of the works shall be confirmed by the S.O. prior to the Contractor mobilising site investigation resources to Site.

G3.2.6 Property of the Board

(a) All the results of the work in the Contract including originals of borehole logs etc., drawings and bench or work sheets together with any samples and cores obtained shall be deemed to be the property of the Board.

G3.2.7 Boring - General Requirements

(a) The boring and drilling rigs shall be of the light cable tool percussion type or rotary hydraulic feed type, skid mounted with a drilling head capable of recovering 100 mm diameter undisturbed samples in approved sample tubes. Wash bore techniques shall not be permitted.

(i) Boring Rigs:

1) Boring rigs shall be capable of completing holes to a depth of 30 metres at a minimum diameter of 150 mm. Boring may be carried out by percussion tools but not by water jets unless permitted otherwise by the S.O..

2) Working Platform:

- The Contractor shall provide such platforms as may be necessary to keep boring and/or drilling rigs level when putting down holes on sloping or waterlogged ground.

3) Advance of Investigation Holes:

- The Contractor shall advance the investigation holes by using rotary drilling or light cable tool percussion methods with casing. Investigation holes shall be cased in any stratum which is friable or not sufficiently strong to stand unsupported, or when directed by the S.O.. The size of the cased hole shall be such that the requirements described in the above and following clauses in respect of sampling, groundwater observation, field tests and records are satisfied. The Contractor shall be solely responsible for ensuring that the casings are inserted in such a manner as to render them recoverable and the Contractor shall have no claim for damage, loss or delay caused by difficulty or failure to recover a casing;
- Casing shall not be removed from any hole nor any filling introduced into a hole without the prior permission of the S.O.. This permission will normally be given as soon as work in the hole is complete. Casings shall gradually be withdrawn and the grout filling shall be kept above the base of the casing during withdrawal;
- Drilling fluid shall normally be clean water, but drilling mud shall be used when directed by the S.O. during the drilling of holes in alluvium and other soft deposits.

4) Depth of Investigation Holes:

- The depths and number of investigation holes shown in the Tender Documents are provisional. The S.O. may vary the location and quantities and it shall be understood that the variation shall not affect the rates in the priced Bill of Quantities.

5) Obstructions:

- It shall be the Contractor's responsibility when boulders or other obstacles are encountered to carry the boring/drilling through or past such obstacles or, if directed, to core the obstacle to determine its size and character. The Contract Rates shall include for chiselling through obstructions.

6) Abandoned Investigation Holes:

- Except with the specific permission of the S.O., the Contractor shall not abandon or complete a drilling or remove any casing or drilling equipment without first affording the S.O. the opportunity of obtaining the position and depth of the drill hole before abandonment or completion, and any other information which he may require. No allowance or payment whatsoever shall be made for any drilling abandoned or completed without compliance with these stipulations. In addition, in order to receive consideration for payment for abandoned drill holes, the Contractor shall furnish the S.O. with complete records and samples for the depths penetrated in the manner prescribed for completed drill holes.

7) Termination of Investigation Holes:

- The termination of each investigation hole shall be decided by the S.O.. The Contractor shall inform the S.O. immediately on reaching the depth directed by the S.O..

G3.2.8 Boring - in Overburden

(a) Boring in overburden (alluvium, slopewash and residual soil) shall be by light cable tool percussion as directed by the S.O. so as to achieve maximum recovery. Rotary drilling in soil may in some circumstances be approved by the S.O. but is not preferable, and in some circumstances water or drilling mud shall be directed for holes which penetrate alluvium.

(b) The diameter of the holes shall be at least 150 mm at the bottom of the hole and over the entire depth of the hole shall be such as to permit the taking of all required types of samples and the removal of all other material in such a way that it can be accurately identified and the levels of each stratum accurately determined to the full depth of the borehole.

(c) In boring through permeable materials, the Contractor shall avoid any unnecessary disturbance to the material and shall ensure that:

(i) The water level in the hole shall be maintained slightly above the water table in the permeable strata;

(ii) The casing shall be left as near the base of the hole as possible; and

(iii) Close fitting tools shall be removed slowly to avoid suction pressure.

(d) When a borehole is being cased in granular or soft fill, the bottom of the casing shall always be maintained within 150 mm of the bottom of the borehole. The casing shall never be in advance of the bottom of a borehole during undisturbed sampling or standard penetration testing.

G3.2.9 Groundwater

(a) Groundwater Observation

(i) During all boring/drilling operations, the Contractor shall have on site an approved electrical water level measuring device. Levels shall be read to an accuracy of ± 10 mm;

(ii) All records of measurements shall contain the date and the time of day on which the measurement is made. Readings shall be taken at the times set out in Clause 3.14.19, and when required by the S.O..

(b) Installation of Standpipes and Piezometers

(i) Standpipe piezometers for monitoring groundwater levels and changes in groundwater level at a particular depth in exploratory holes shall be installed as instructed by the S.O.. They shall be as described in Clause 3.14.16 of this Specification and all dimensions and depths shall be recorded;

(ii) Daily readings of depths to water in piezometers shall be made by the Contractor with an approved instrument during the fieldwork period or as directed by the S.O..

G3.2.10 Filling and Sealing of Investigation Holes

(a) All investigation holes shall be backfilled with grout. Grout for backfilling holes shall consist of a mud formed by mixing one part by weight of bentonite with 10 parts of water, to which shall be added two parts by weight of cement after the bentonite and water have been thoroughly mixed. The grout shall be fed into the bottom of the hole by a tremie pipe, the end of which shall always be below the grout water interface whilst grouting is being carried out. Alternatively, holes may be backfilled with purpose made pellets of bentonite or bentonite/cement, provided they are of a size which, in the opinion of the S.O. is compatible with the size of hole and any piezometer tubes to be installed in it, and provided they are poured in carefully so that they do not jam above the bottom of the hole.

(b) Grout backfill shall be stopped 1 metre below the original ground level and the remaining depth of the bore shall be filled with dry topsoil rammed hard except that hard surfaces shall be backfilled and reinstated to the standard of the surrounding ground to the satisfaction of the S.O..

(c) Casing and timbering shall not be removed from any hole and filling shall not be introduced into a hole until permission is given by the S.O.. This permission will normally be given as soon as work in the hole is completed. Casing shall be withdrawn gradually and the grout backfill shall be kept above the bottom of the casing during withdrawal.

(d) The Contractor shall avoid leaving a hole overnight after he has begun to withdraw the casing and before backfilling.

(e) Where a borehole is to be continued as a drill hole, drilling casing shall be inserted before the boring casing is withdrawn but no grouting will be required until the drilling casing is finally withdrawn.

G3.2.11 In-situ Testing - General

(a) General

(i) The Contractor shall perform in situ tests described in investigation holes or without holes as appropriate.

(ii) In situ testing in investigation holes may include:

1) Standard penetration tests (SPT);

2) Vane shear tests; and

3) Approval of Test Equipment.

(b) All the proposed testing equipment shall be subject to approval by the S.O. with respect to its general condition, suitability for the intended use, calibration where appropriate and compliance with the specified standards.

G3.2.12 In-situ Testing - Standard Penetration Tests

(a) General

(i) In predominantly granular soils or where directed by the S.O., standard penetration tests (SPT) shall be carried out by the Contractor.

(b) Equipment

(i) The equipment shall be in good working condition and shall comply with BS EN 1997-2. The hammer shall be an automatic trip hammer.

(c) Test Method

(i) The test method shall be according to BS EN 1997-2. A split barrel sampler shall be used except that a cone tool may be used in gravels with the approval of the S.O.. When 50 blows on the sampler results in less than 150 mm penetration, the test shall be discontinued and the actual penetration recorded;

(ii) The whole of the material removed from the SPT split tube shall be immediately sealed to preserve its natural moisture content. If more than one type of material is represented, each soil type shall be placed in a separate container, taking care to record the relative position of each type of material. When a satisfactory sample cannot be obtained from the split barrel of the sampler, or when a cone tool has been used, a representative disturbed sample with a mass of not less than 0.5 kg shall be obtained by some other approved technique;

(iii) A small disturbed sample shall be taken from each soil type in the sampler core. The water level at the time of the test shall also be recorded. Where a solid cone is used or where no soil is recovered in the sampler a disturbed sample shall be obtained from the position of the test; and

(iv) Unless otherwise directed by the S.O. the frequency of the in-situ testing shall be at 1.0 m intervals for the first 5 m then at 1.5 m intervals thereafter.

(d) Results

(i) Field results, giving the number of blows required for the seating drive and per 75 mm increment of penetration, shall be recorded on the borehole logs.

G3.2.13 In-situ Testing - Vane Shear Tests

(a) General

(i) Vane shear tests shall be performed where directed by the S.O., but shall generally only be performed in soft alluvial clays and silts.

(b) Equipment

(i) The equipment shall satisfy the requirements of BS EN 1997-2, except where this Specification indicates otherwise in which case the Specification shall be followed. The apparatus shall enable the determination of rod friction, undisturbed soil shear resistance and remoulded soil shear resistance;

(ii) The Contractor shall supply three standard sizes of rectangular vanes (not tapered ends) measuring 51 mm, 76 mm and 102 mm in diameter. Rods used for applying torque to the vane shall be as new, free of bends and shall not be used for other drilling or testing functions, specifically SPT tests. Rod stabilisers shall be provided at intervals of 3 m. Provision shall be made for anchoring the shear vane body to prevent counter rotation. The Contractor shall provide a recent certificate of calibration for the load measuring device.

(c) Test Procedure

(i) After installation of the vane of a size and at a depth directed by the S.O., the Contractor shall allow 30 minutes before commencing the test. The vane shall be rotated at a constant rate of between 5 and 6 degrees per minute and recordings of load versus angular rotation shall be made at intervals of 5 degrees angular rotation. After the soil has been sheared initially, the vane shall be rotated rapidly for eight revolutions and the remoulded shear resistance measured after a delay of 5 minutes. Measurements of remoulded shear resistance shall be made over an angular rotation of at least 90 degrees.

(d) Results

(i) Field result sheets shall include the following details:

1) Project Title and location description;

2) Hole number;

3) Test number;

4) Depth of test;

5) Dimension of vane;

6) A graph showing torque versus angular rotation;

7) Calculations showing peak and residual shear strength; and

8) Date of test.

G3.2.14 Soil Instrumentation and Monitoring

(a) General

(i) The Contractor shall install geotechnical instrumentation to monitor the long term groundwater conditions at the site.

(b) Scope of Work

(i) Installation of standpipe piezometers at the direction of the S.O.; and

(ii) Monitoring of the long term groundwater conditions beneath the raw water siphon and landfill embankments.

(c) Instrumentation - General

(i) Instrumentation shall be installed to measure the groundwater conditions beneath the embankments to be constructed under this Contract. Instrumentation is part of the permanent works and shall remain operational both during and after the site investigation component of the Works. The Contractor shall maintain, monitor and report on all instrumentation until the end of the Defects Liability Period. In particular the standpipes and piezometers shall remain in place whilst the embankment construction is completed, with the vertical legs of the standpipes and piezometers extended to suit.

(ii) The Contractor shall be responsible for and follow the instructions of the manufacturer and the requirements of this Specification in the installation and testing of all measuring instruments which shall be carried out under the direct supervision of the S.O. or his representative. The Contractor shall inform the S.O. at least 2 days prior to undertaking installation of the equipment.

(d) Instruments - Protection and Maintenance

(i) The Contractor shall take necessary precautions to protect the instruments and maintain the instruments in good working order after commissioning. For all instruments which project through and above ground, special precautions shall be taken to provide protection from vehicles and plant including substantial and readily visible barriers at a distance of 750 mm around each instrument. The construction of the barriers shall be approved by the S.O.. Damaged instruments shall be replaced or repaired by the Contractor at his own expense within three days.

(e) Stabilising Electronic Readout Devices

(i) All electronic readout devices and transducers shall be shaded from direct sunlight during use. Probes which are used inside access tubes shall be placed inside the tube and allowed to come to a stable temperature for at least 10 minutes before use. Zero or starting values shall only be taken once temperature stabilisation is complete.

(f) Personnel

(i) A Senior Instrumentation Technician with at least 2 years' experience using the equipment to be installed shall be appointed;

(ii) The Senior Instrumentation Technician shall be responsible for the overall planning, implementation and monitoring of the instruments. He shall be on site throughout installation, commissioning and all monitoring. He shall be involved in the remaining monitoring on a basis to be agreed with the S.O.. He shall ensure that all instrumentation works and monitoring is either carried out by or supervised by a suitably experienced technician; and

(iii) The Contractor shall submit names and curricula vitae of personnel to carry out the instrumentation and monitoring and a programme of their attendance for the approval of the S.O.. Deviation from the approved programme of attendance or the requirements given in this Clause shall only be permitted with the approval of the S.O..

(g) Grouting of Boreholes

(i) For all instruments placed in boreholes, grouting is required of part or all of the borehole during installation, the grout shall be a bentonite: cement mixture with sufficient water to achieve a pumpable mix. The proportions of the mix shall be such as to imitate as closely as possible the strength or consistency of the natural soils present. The Contractor shall conduct trials on different mixes of bentonite: cement to ascertain the relationship with strength. Specimens shall be cured and stored, then tested in undrained triaxial compression tests after 1 day, 7 days, and 1 month. Three specimens shall be tested on each occasion, and the sources of bentonite and cement shall be the same as used for eventual installation. On the basis of these trails, the S.O. shall decide on the bentonite: cement proportions to be used, which may be varied depending on the application.

(ii) Grout shall be poured or pumped into boreholes using a tremie pipe.

(h) Labelling and Marking of Instruments

(i) All instruments shall be labelled with their reference number at the location where readings or measurements are taken. The labelling shall be permanent using a method or material to be agreed with the S.O..

(i) Survey Equipment

(i) All surveying equipment used in conjunction with the monitoring of instrumentation, including measuring tapes, levels, theodolites and all electronic survey equipment shall be maintained and calibrated as required by the manufacturers, SISIR, and good surveying practice. Levels shall be used to check the horizontality of the line of sight every four weeks. SISIR calibration certificates shall be provided for all survey equipment which the Contractor proposes to use.

(j) Instrumentation Equipment References

(i) Only the geotechnical instrumentation and equipment specified shall be installed.

(ii) Equipment from other manufacturers may be used provided that it meets the requirements of this Specification and provides the same facilities. Details of alternative equipment shall be submitted to the S.O. for approval. If the S.O. considers it necessary, demonstrations shall be arranged by the Contractor. Any supplier of geotechnical instrumentation must demonstrate that they operate an adequate system of product quality assurance.

G3.2.15 Standpipe Piezometers

(a) Scope

(i) Standpipe piezometers shall be used to give a measurement of water pressure at a specific depth within a soil profile. They are generally used in soil of medium to high permeability. The Contractor shall install standpipe piezometers at locations, and with depths and details as specified by the S.O..

(b) Equipment and Installation

(i) The piezometer tip shall consist of a porous ceramic or plastic element not less than 150 mm long with a diameter not less than 30 mm, and shall be protected at each end by unplasticised polyvinylchoride (uPVC) fittings. The filter element shall have a pore diameter of the order of 60 microns and a permeability of the order of 0.0003 m/s. The tubes shall be jointed together and to the porous element with threaded couplings, PTFE tape and glue in such a manner that the joints remain leak-proof under the anticipated head of water;

(ii) The standpipe piezometer shall be installed in a 100 mm diameter borehole. The sand filter surrounding the porous element shall be clean and fall wholly between the limits of grading 600 and 1200 microns. The Contractor's arrangements shall ensure that no sand adheres to the soil in the sides of the unlined borehole. Where there is water in the borehole the Contractor shall allow sufficient time for all the sand to settle. The final elevation of the top of this sand shall be recorded. The porous element shall be placed in the hole and the remaining sand filter shall then be added as described above. The final elevation of the top of the sand filter shall be measured by a flat-ended sounding rod; and

(iii) Seals consisting of bentonite pellets shall be placed above, and if necessary, below the sand filter. The remainder of the hole shall be filled with a bentonite: cement grout, and the top part with a concrete plug. A protective cover shall be set into concrete with caps and air vents.

(c) Commissioning

(i) Before taking initial readings, the Contractor shall carry out a simple falling head test by raising the water level 1.5 m above the static level, using an extension pipe if necessary, and measuring the water level over a 30 minute period.

(d) Method of Monitoring

(i) The depth to water in standpipe piezometers shall be measured using a dipmeter. The dipmeter shall be of the electric type but simple metal probes attached to nylon cord may be used with the approval of the S.O. for shallow depths.

G3.2.16 Instrumentation Records

(a) Commissioning and Base Readings

(i) After installation, the functioning of each instrument shall be demonstrated to the S.O., including the recording of measured values using the appropriate readout device. As part of the commissioning three sets of readings shall be taken and compared.

(b) General Information on All Records

(i) All records of instrumentation, either installation details, routine readings or monthly summaries, shall contain the following information:

1) Project name;

2) Contract name and number;

3) Instrument reference number and type;

4) Dates of installation, reading or summary;

5) Times of installation or reading;

6) Personnel responsible; and

7) Relevant comments or remarks.

(c) Installation Records

(i) The Contractor shall prepare an installation record sheet for each instrument installed. The format of the sheet shall be prepared by the Contractor and submitted to the S.O. for approval at least one week before installation commences. The record sheet shall include the following information in addition to the general information required:

1) Existing ground level at the time of installation;

2) Planned location in plan and elevation;

3) Planned lengths, widths, diameters, depth and volumes of backfill;

4) Plant and equipment used, including diameter and depth of any drill casing used;

5) Type of backfill used;

6) As-built location in plan and elevation;

7) As-built lengths, widths, diameters, depths and volumes of backfill;

8) Weather conditions;

9) A space for notes, including problems encountered, delays, unusual features of the installation, and any events that may have a bearing on instrument behaviour;

10) A record of commissioning information and readings; and

11) Any colour coding used.

(ii) The Contractor shall submit to the S.O. the specified number of copies of each installation report within one working day of completion of the installation, including taking of base readings.

(d) Readings

(i) On each occasion that readings are taken from an instrument the measured values shall be recorded on a record sheet. The format of the record sheet for each type of instrument shall be prepared by the Contractor and submitted to the S.O. for approval at least one week before readings commence. Details of information and values to be stored on each record sheet in addition to the general information required are given below:

Instrument	Data required
Standpipes piezometers	-depth to water from top of piezometers tube (m)
	-reduced level of top of tube (mLSD)
	-reduced level of ground adjacent to standpipe (mLSD)
	-water head (mLSD)
	-change in water head relative to base readings (m)

(ii) The Contractor shall submit to the S.O. the specified number of copies of each record sheet with necessary listings within one working day of taking the readings.

(e) Frequency of Readings

(i) The Contractor shall monitor water levels in standpipes and standpipe piezometers once daily for the first seven days after the piezometers are installed. Thereafter water levels shall be monitored at seven day intervals or as directed by the S.O. During periods of continued rainfall, the water levels shall be monitored once daily for a duration as directed by the S.O..

(f) Anomalous Readings

(i) Whenever sets of data are measured, they shall be compared to previous sets of data. If anomalous readings are present which differ from the expected value or trend, then further readings shall be taken immediately and the S.O. shall be informed. If the anomalous values persist, then the S.O. shall be informed and an investigation shall be carried out to find the reasons for the anomalous readings.

G3.2.17 Daily Field Records

(a) Each day during the site investigation component of the Work on the Site, the Contractor shall hand to the S.O. the original and fair copies in duplicate of the records of the previous day's work containing the following information in respect of each hole where work was in progress:

(i) Scheme number and site name;

(ii) Number, type and size of the hole;

(iii) Date and hours worked on the Site;

(iv) Brief description of the weather including the amount of rainfall;

(v) Total depth of hole at the beginning and end of each shift;

(vi) Make and type of machine in use;

(vii) The water levels in accordance with clause G3.2.18 below and the depths at which water inflows were encountered;

(viii) The approximate quantity of water poured into a borehole and the time(s) when it was done;

(ix) Diameter of the hole and depths of any reduction in diameter;

(x) The length of hole for which casing was used and the diameter of such casing;

(xi) Reduced level of ground;

(xii) A full geotechnical description of each stratum encountered;

(xiii) Depth below ground of each change of stratum;

(xiv) Data obtained during in situ tests, including the water level in the hole during the test;

(xv) Details of backfilling and grouting;

(xvi) Details of any instrumentation installed;

(xvii) Details of delays and breakdowns; and

(xviii) Any other relevant information and details of any other operation for which the Contractor may require additional payment.

G3.2.18 Water Level Records

(a) Readings of water levels in all holes where work is in progress shall be taken with an electrically operated sounder and recorded in the daily field records and logs at the following times:

(i) Before work commences in the morning;

(ii) Before the mid-day break;

(iii) After the mid-day break;

(iv) After work has finished in the evening, both before and after water (if any) is added to stabilise the ¬hole;

(v) When a hole has been completed;

(vi) Immediately prior to backfilling a hole;

(vii) At the time of undisturbed sampling, standard penetration tests and other in situ tests; and

(viii) When the S.O. requires additional measurements.

(b) The level of the bottom of the hole and the bottom of the casing, if any, shall be measured and recorded at the same time as each water level reading.

(c) The times when water levels are measured shall also be recorded. If, at any time, the level of the water in a hole fluctuates, a record shall be kept of the fluctuation. If the hole "makes" or "loses" water the S.O. shall be informed immediately.

(d) Any addition of water to assist the advance of a hole shall be recorded and any extraordinary odour or colour of the water and any other unusual circumstances shall be reported. Water shall not be added or removed from a hole when in the opinion of the S.O. such action might adversely affect undisturbed sampling and the results of in situ tests.

(e) When groundwater is encountered the depth below ground level of the point of entry shall be recorded and operations stopped and the depth to water level recorded at 5 minute intervals until no further rise is observed. However, if at the end of a period of 20 minutes the water level is still rising, unless otherwise instructed by the S.O., this shall be recorded and the exploratory hole continued. On each occasion when groundwater is encountered the depth of the hole, the depth of any casing and the time shall also be recorded.

(f) After expiry of the first 20 minutes any time during which work is stopped on the order of the S.O. in order to allow the water level to stabilise itself shall qualify as `authorised standing time'.

G3.2.19 Exploratory Hole Logs

(a) Logs of boreholes shall be provided in the form of the specimen sheet included at the end of the Specification. The format of all records shall be subject to the approval of the S.O.. Two draft copies shall be submitted to the S.O. within 5 working days of the completion of the exploratory hole to which they refer and shall contain the following information:

(i) Scheme number and site name;

(ii) Contractor's name;

(iii) Exploratory hole reference number, location (grid reference or coordinates) and diameter or size;

(iv) Dates of exploration referred to the depth at the end of each working day or shift;

(v) Equipment in use;

(vi) Data on the stability of the exploratory hole and details of casing used related to progress;

(vii) Ground level referenced to Land Survey Datum;

(viii) Depth to each change of stratum;

(ix) Records of groundwater; depth encountered, height to which water rose, rate of inflow, depth cut off by casing or shoring;

(x) The depth and details of all in-situ tests;

(xi) Elevation of each stratum relative to Land Survey Datum;

(xii) Symbolic legend of strata encountered in accordance with the standard legends;

(xiii) Details of backfilling and grouting;

(xiv) Details of any instrumentation installed; and

(xv) Systematic engineering description of each stratum in accordance with the standard format.

G3.2.20 Clarity of Records

(a) Particular attention shall be given on Site to ensure that all investigation records are written down without delay and that they are accurate, complete, clear and legible. Blue ball point pen shall be used for hand written work as proof of original copy. No correction pen to be used and changes made shall be counter singed Faint photocopies will not be acceptable. If data is omitted for any reason, the reason must be stated.

G3.2.21 Reports

(a) The Contractor's Final Report shall contain factual information and analyses of laboratory data as required by this Specification. The report shall consist of:

(i) A general description of the work carried out; the numbers and types of exploratory holes, plant and equipment used; a factual summary of general geology and subsurface conditions;

(ii) A map showing the location of the Site, and plans indicating the location of field tests;

(iii) Borehole logs;

(iv) Detailed results of all field tests; and

(v) Instrumentation details and results.

(b) The Contractor shall submit to the S.O. three (3) copies of a Draft Report containing all the information gained during the field investigations. The Draft Report shall be submitted as soon as possible and not later than two (2) weeks after the S.O. has notified the Contractor that no further site investigation field work will be required. The S.O. after studying the Draft Report may require the "presentation" of the report to be altered. "Presentation" shall be taken to include the size of the report, drawings, title on cover, quality of cover and binding. The reports shall be bound and shall be on ISO paper size A4. Drawings larger than A3 shall be folded and included in bound wallets.

(c) After the S.O. is satisfied with the Draft Report, the Contractor shall make all necessary amendments and shall prepare ten (10) copies of the Final Report and softcopy in AGS format.

END OF SECTION